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nd processing speed, show age[1]. Pathological conditions iated with progressive neuro- er impact cognitive function. In addition to dementia, visual impairment is also one of the most common disorders among older populations[2]. Studies have shown that visual impairment is associated with an increased risk of developing dementia and cognitive decline[2,3]. Individuals with co-existing cognitive and visual impairments are at a higher risk of disability, which can further reduce independence in everyday activities[4].

Management and treatment strategies for cognitive impairment include both pharmacological and non-pharmacological methods. Pharmacological interventions typically involve the use of medications designed to alleviate cognitive symptoms and correct specific neurochemical imbalances. Although these medications may provide some degree of symptomatic relief, they can lead to various side effects, and their long-term effectiveness in slowing disease progression is often limited[5]. On the other hand, non-pharmacological approaches encompass a wide range of techniques, including cognitive training, cognitive stimulation, physical activity, dietary modifications, and social engagement, all aimed at improving overall brain health, reducing risk factors, and promoting neuroplasticity[6]. For example, adopting a lifestyle that includes regular physical exercise and a balanced diet can enhance cerebral blood flow and reduce the risk of further cognitive decline[7]. Cognitive training, which usually targets specific cognitive domains, consists of structured exercises and activities intended to strengthen particular cognitive skills, thereby enhancing memory and executive function[8]. Additionally, participation in social activities can help individuals manage cognitive difficulties[9]. Non-pharmacological interventions are particularly valuable in enabling individuals to take an active role in managing their condition, potentially resulting in better long-term outcomes. Moreover, these interventions do not carry the risks and side effects associated with pharmacological treatments, making them a safer and more sustainable option.

Computerized Cognitive Training (CCT) is one of the non-pharmacological interventions used in managing cognitive decline. CCT is defined as any form of cognitive training involving the use of an individual electronic device[10]. It consists of guided drill-and-practice tasks designed to target one or increasingly applied in clinical contexts, as it is a safe, inexpensive, and beneficial method to improve cognitive functions in individuals with dementia or those at risk of developing the condition[12]. A review showed that CCT can improve cognitive domains (*e.g.*, attention, processing speed, and visuospatial memory) and everyday functioning, with small to moderate effects in older adults without health conditions affecting cognition[13]. In individuals with mild to moderate dementia, a systematic review of 33 studies found that, compared to no intervention, CCT can have positive effects on global cognition and verbal fluency[14]. There is also evidence suggesting that CCT may benefit attention and memory in individuals with mild cognitive impairment[6,15].

There is a limited number of studies investigating the effects of CCT on cognition in individuals with both cognitive and visual impairments. One apparent reason is that CCT often requires intact vision, and training materials typically involve visual stimuli and cues[10,12]. This study reported an auditory CTT protocol using commercially available software in a group of participants with both cognitive and visual impairments. Due to participants' visual limitations, two training tasks that do not require vision and can be completed using auditory input were selected. Individual sessions and a supervised approach were employed, as these have been reported to produce greater cognitive effects[16]. The selected tasks specifically targeted selective and focused attention, and it was hypothesized that improvement in attentional performance would be observed following the training.

METHODS

Participants

Sixteen participants were recruited from residential facilities for visually impaired older adults. Selection criteria were as follows:

- aged 65 or above;
- reported cognitive complaints (either self-reported or reported by family members);
- visual acuity equivalent to 20/200 or worse in the better eye, indicating moderate to severe visual impairment or blindness.

Participants were excluded if they met any of the following criteria:

on;

tellectual disability, central nervous tion, brain tumor, traumatic brain cant psychiatric disorders (such as major depression or schizophrenia), substance abuse, or other medical conditions (excluding dementia) that may interfere with cognition;

- communication barriers, including but not limited to deafness or other significant language impairments.

Participants were recruited through a convenience sampling method. Staff at the residential facilities reviewed medical charts of potential participants to ensure eligibility. Eight participants had a formal diagnosis of dementia. All outcome measures were conducted by occupational therapists. Ethics approval for this study was obtained from the author's affiliated college. Written informed consent was obtained from participants or their family members prior to the start of the study.

Assessments

Neuropsychological assessments included the Cantonese version of the Mini-Mental State Examination (CMMSE)[17], the digit span forward test[18-21], the Chinese version of the Verbal Learning Test (CVVLT)[22], and the memory subtest of the Neurobehavioral Cognitive Status Examination (NCSE)[23]. CMMSE (maximum score of 30) was used to screen the overall cognitive function and was administered only at pre-test. The digit span forward test (maximum score of 9) was used to assess attention and short-term memory[19-21]. CVVLT, which measures working and episodic memory, consisted of nine two-character Chinese nouns with four learning trials (maximum score of 9 per trial; total maximum score of 36) and one 10-minute delayed recall trial (maximum score of 9). The memory subtest of the NCSE required participants to memorize and later recall four two-character Chinese nouns (maximum score of 12). Except for the CMMSE, all other assessments were administered within two days before and after the training sessions.

Intervention

A total of eight individual sessions were conducted once to twice a week (every 5 to 6 days) over a 6-week period for each participant. Each session lasted approximately 45 minutes, with 30 minutes dedicated to actual training. Two cognitive training modules (CogniPlus,

RESULTS

Two participants did not complete the training sessions. The remaining 14 participants were all females (mean age: 85.4±11.8 years; years of education: 3.4±4.4 years). Their mean CMMSE score was 18.1±4.7 (range: 12–26). Paired-samples *t*-tests were performed to compare the pre- and post-training mean assessment scores. Results are shown in Table 1. Significant differences were found in four assessments (digit span forward test, CVVLT first trial, CVVLT total, and NCSE-memory). No significant difference was found in the CVVLT delayed recall trial. Analysis using the Wilcoxon signed-rank test showed similar results. Additional repeated measures analyses

p. The two subgroups did not of education (younger *vs*. older cores (20.8 *vs*. 15.8), and both with a dementia diagnosis.

DISCUSSION

To our knowledge, this was the first trial to include a group of aged participants with visual impairment for CCT. Results showed improvements in the digit span forward test, CVVLT first trial, CVVLT total, and NCSE-memory subtest scores. These findings provide preliminary evidence that auditory-based CCT may be beneficial for attention and memory in individuals with both cognitive and visual impairment. A previous large-scale study of auditory-based CCT in community-dwelling older adults aged ≥60 years at risk of progressive cognitive decline[24]. Their CTT aimed to improve speed and accuracy of auditory information processing lasted for one hour each day, five days a week, over a period of eight weeks. The participants in the cognitive training group showed improvements in sustained attention and working memory post training and sustained at 6 month[24]. No improvements were found in the active control or no-contact control groups. These findings support the notion of neuroplasticity, in which behavioral changes may be achieved through targeted cognitive training[25,26],

Table 1. Pretest-Posttest Comparison. Data are shown as mean±standard deviation.

	Pretest (N=14)	Posttest (N=14)	*p*-value	Cohen's d	95%CI
Digit span forward test	5.14±1.56	6.14±1.99	<0.001	-1.28	-1.45 to -0.54
CVVLT first trial	2.00±0.78	3.14±1.79	0.017	-0.73	-2.04 to -0.24
CVVLT total	11.79±4.59	13.57±6.35	0.008	-0.84	-3.14 to -0.57
CVVLT delayed recall	1.00±1.96	1.21±2.29	0.19	-0.37	-0.54 to 0.12
NCSE—memory	9.57±1.87	10.64±0.93	0.05	-0.58	-2.14 to 0

Abbreviations: CVVLT, Chinese Verbal Learning Test; NCSE, Neurobehavioral Cognitive Status Examination; 95%CI, 95% confidence interval.

Table 2. Additional tests for age and education groups. Results from interaction effects.

	Age x time (F value)	*p*-value	Education x time (F value)	*p*-value
Digit span forward test	2.0	0.183	0.444	0.518
CVVLT first trial	5.76	0.034*	0.109	0.747
CVVLT total	4.83	0.048*	0.928	0.355
CVVLT delayed recall	2.08	0.175	2.08	0.175
NCSE—memory	1.014	0.334	2.858	0.117

Abbreviations: CVVLT, Chinese Verbal Learning Test; NCSE, Neurobehavioral Cognitive Status Examination.

participants. Participants could practice unlimited times until successful repetition of all four nouns twice. Besides, if the participant could not recall a noun in the NCSE–memory subtest, categorical cues and choices were provided, and they could still get partial marks for items recalled with cues. This indicated that these participants could benefit and might need more support at the retrieval stage to show their improvements.

There were several limitations in this study. The sample size was small (especially for the sub-group analysis), and participants were more heterogenous (baseline cognitive performance differed). There was an absence of a control group, which made the results more prone to type 1 error. In addition, follow-up effects were not assessed. Nevertheless, the results of this study provided preliminary support for the efficacy of an auditory-based cognitive training program in people with both cognitive and visual impairment. In people with higher computer proficiency, the program could be adopted to allow web or mobile app-based training so that they could self-administer at home at their own time. Android or Apple devices already have accessibility functions, such as TalkBack or VoiceOver, embedded. Though the integration and adaptation would require time and careful considerations, developers such as UnoBrain has seen the need and have a goal of producing nine accessible brain games for the blind and partially sighted. Further studies that include a more homogenous sample and a control group are warranted. Investigations can also consider adding a follow-up test, *e.g.*, at 1-month post-training or longer, to evaluate whether there may be long-term effects. Future studies may consider exploring the effects of multimodal programs such as CCT with physical exercises or brain stimulation and investigate whether there may be additive beneficial effects.

AUTHORS' CONTRIBUTIONS

Conceptualization: MCCK; Data curation: ATSC; Formal analysis: MCCK; Validation: MCCK, ATSC; Visualization: MCCK; Writing — original draft: MCCK; Writing — review & editing: MCCK, ATSC; Investigation: MCCK, ATSC; Project administration: MCCK, ATSC; Resources: MCCK, ATSC; Funding acquisition: MCCK; Supervision: MCCK.

DATA AVAILABILITY STATEMENT

The participants of this study did not give written consent for their data to be shared publicly, so due to the sensitive